

BRIAN KIPRUTO.

bryanrutto4@gmail.com | +254 703 480806 | www.byteanza.com

 www.linkedin.com/in/brian-kipruto

 https://github.com/Ruto20

SUMMARY

A highly motivated Physicist with a strong foundation in astrophysics and a proven track record of applying scientific principles to develop innovative technological solutions. Possesses expertise in robotics, sensor technology, and data analysis, demonstrated through leading projects in radiation assessment and smart agriculture. Seeking to leverage this experience and advanced knowledge in nuclear physics to contribute to the development of sustainable energy and radiation protection.

WORK EXPERIENCE

Autonomous Radiation Mapping Robot - ByteAnza Research

Jan 2025-present

<https://byteanza.com/research/portfolio-details.html>

Leading the design and development of a semi-autonomous robotic system for environmental radiation mapping in Marsabit County, Kenya. This project directly applies principles of nuclear physics and radiation detection to address the critical need for improved environmental assessment in regions potentially affected by radiological contamination.

- The system integrates a 6WD robotic platform with a micro-controller, On-Board Computer, Geiger-Müller tube, GPS, and IMU to acquire precise spatial measurements of radiation levels.
- Developed software for data processing and map generation, enabling a deeper understanding of radiation dispersion and its implications for environmental and public health, and providing valuable insights for developing effective monitoring and mitigation strategies for radiological hazards.

STEM Educator, The Bytelab. (Part time)

Nov 2023 - Present

<http://www.byteanza.com/bytelab/>

The ByteLab aims to foster a love for STEM in children from a young age through hands-on Experiential Learning.

- Led the planning, development, and execution of dynamic STEM education programs for young learners, encompassing electronics, coding, robotics, and IoT.
- Managed program delivery, including instructional sessions on microcontroller programming (Arduino), circuit design, sensor integration, and automation, guiding students through hands-on project development and prototyping.
- Developed scientific and technological curriculum and learning resources, ensuring engaging and effective content delivery.
- Mentored students, providing career guidance and fostering interest in STEM fields.

Scientific Researcher, ByteAnza.

Jan 2023 - Nov 2023

www.Byteanza.com

- Led the research, development, and implementation of Bytegrow, an innovative project delivering smart, autonomous agricultural solutions for small- and medium-scale farmers.
- Managed the project lifecycle, from prototype development (a fogponics system for water conservation using Arduino and ESP8266 microcontrollers) to data collection and cloud storage integration.
- Achieved project milestones, resulting in recognition as a top 10 finalist in the AYUTE Kenya Challenge 2023 and a winning entry at the KNBS Hackathon 2023.

Machine Learning Research Intern & Project team member, University of Konstanz. Germany.

May 2022 - Aug 2022

- Contributed to a research project focused on machine learning applications for particulate structure detection in colloidal solutions.
- Conducted data analytics and visualization using Ovito.
- Participated in the development and implementation of neural networks for studying and predicting particulate structures.

EDUCATION

MSc Nuclear Science

Aug 2025 - Aug 2027

University of Nairobi.

My research revolves around using machine learning to build a virtual prototyping tool for rapid performance prediction of flexible Graded-Z shielding material composites.

BSc Astrophysics

Sep 2019 - Sep 2023

University of Nairobi

- Second Class Honours.

VOLUNTEER WORK

Nuclear Society of Kenya (Member)

Jan 2025 - Present

My volunteer work with the Nuclear Society of Kenya allows me to engage in public outreach, professional development, and policy advocacy, which aligns directly with my long-term goal of fostering a strong and responsible nuclear ecosystem in Africa. This experience further motivates my pursuit of advanced studies to acquire the expertise necessary to lead in this crucial area.

Volunteer Researcher, International Astronomical Asteroid search campaign

Oct 2021 - Nov 2021

- Applied data analysis techniques to telescope data to identify and locate potential asteroids using Astrometrica software.

Moon Village Association

Mar 2021 - Nov 2022

- Served as Deputy Coordinator at MVA-PESC Kenya (Moon Village Association - Participation of Emerging Space Countries) in conjunction with the Kenya Space Agency.

FEATURES & CERTIFICATIONS

The African Physics Newsletter.

Website: <https://www.aps.org/publications/african-physics-newsletter>

Got Featured in the July 2024 issue of the African Physics Newsletter for my works at the Bytelab. Where curiosity meets innovation.

Certified Space Professional, Teaching Science & Technology, Inc. (TSTI)

Website: <https://www.tsti.net/>

The Space Mission Fundamentals Course by TSTI Inc., in collaboration with the Kenya Space Agency, provides a comprehensive overview of space systems, missions, and operations. It covers key topics such as orbital mechanics, space environment, mission management, spacecraft subsystems, launch vehicles, and human spaceflight challenges. Designed for engineers, scientists, and professionals new to the space industry, the course equips learners with the ability to analyze space missions, understand trade-offs in system design, and apply space concepts to real-world problems. Through practical exercises, participants gain hands-on experience in mission planning and spacecraft design.

Qubit by Qubit's Introduction to Quantum Computing

Website: gg@the-cs.org

An introductory course in quantum computing. The aim is to expose students to this emerging field while providing them with real-world programming skills for quantum computers. In the course, We learnt about the foundational concepts in quantum computation, including superposition, entanglement, superdense coding, and explored quantum applications.

Citizen Digital

<https://www.citizen.digital/tech/the-bytelab-where-curiosity-meets-innovation-n361062>

"The future of technological innovation lies at the intersection of intelligent systems, sustainability, and global collaboration....."

LANGUAGE SKILLS

Mother tongue(s):Swahili

English- IELTS

LISTENING: 8.5 READING: 8.0 WRITING: 7.0 SPEAKING: 7.5 Overall : 8.0 (C1)

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

German

Horen (Listening) **A2**

Lesen(reading) **A2**

Schreiben(Writing)A2

Sprechen (speaking) A2

Levels: A1 and A2: Basic & Elementary level; B1 and B2: Intermediate & Upper intermediate; C1 and C2: Advanced & near-native level.

DIGITAL SKILLS

- Programming Languages: C++, Arduino & Python
- Software: GEANT 4 (Actively Learning), QT, & Slack openMC(Actively learning)
- Research & Analysis: Physics Research, Qualitative & Quantitative analysis, data analysis & Visualization.
- Other: Design Thinking, AI, IOT, Project management & Teamwork